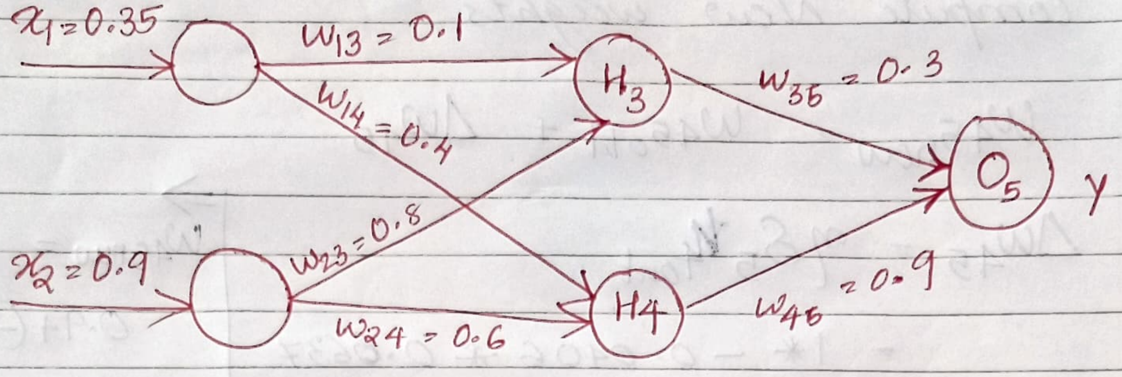

$W_{new} = W_{old} + \Delta w$ Δw – change in weight
where
$\Delta w_{ji} = \eta \delta_j O_i$ ie, derivative of loss w.r.t weight

and $\delta_j = O_j(1-O_j)(T-O_j)$ if j is an output unit

and $\delta_j = O_j(1-O_j) \sum_k \delta_k W_{kj}$ if j is a hidden unit

$\Rightarrow$ Loss/Error = Target output – Predicted output

For Backward pass :-
Compute $\delta_3, \delta_4, \delta_5$

for output unit : $\delta_5 = Y(1-Y)(\text{Target} - Y)$

$= 0.69(1-0.69)(0.5-0.69)$

$= \underline{\underline{-0.0406}}$

for hidden units :-

$\delta_3 = h_{3out}(1-h_{3out})\,\delta_5 w_{35}$

$= 0.68(1-0.68)(-0.0406 * 0.3)$

$= \underline{\underline{-0.00265}}$

$\delta_4 = h_{4out}(1-h_{4out})\,\delta_5 w_{45}$

$= 0.6637(1-0.6637)(-0.0406 * 0.9)$

$= \underline{\underline{-0.0082}}$

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Compute New weights.

$w_{45new} = w_{45old} + \Delta w_{45}$

$\Delta w_{45} = \eta \delta_5 h_{4out}$

$= 1 * -0.0406 * 0.6637$

$= -0.0269$

$\Rightarrow w_{45new} =$

$0.9 + (-0.0269)$

$= 0.8731$

$w_{35new} = w_{35old} + \Delta w_{35}$

$\Delta w_{35} = \eta \delta_5 h_{3out} = 1 * -0.0406 * 0.68$

$= -0.027608$

$\Rightarrow w_{35New} = 0.3 + (-0.027608)$

$= 0.2724$

$w_{14New} = w_{14old} + \Delta w_{14}$

$\Delta w_{14} = \eta \delta_4 x_1 = 1 * -0.0082 * 0.35$

$= -0.00287$

$\Rightarrow w_{14new} = 0.4 + (-0.00287) = 0.3971$

$w_{23new} = w_{23old} + \Delta w_{23}$

$\Delta w_{23} = \eta \delta_3 x_2 = 1 * -0.00265 * 0.9$

$= -0.002385$

$w_{23New} = 0.8 + (-0.002385)$

$= 0.7976$